

AMEL OSMAN

Data Science Graduate Student

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SUMMARY

Aspiring Data Scientist with an M.S. in Data Science from Willamette University. Experienced in the full data pipeline—from building robust databases and pipelines to deploying predictive models—with a focus on translating complex data into actionable insights for social impact.

TECHNICAL SKILLS

- Programming Languages: Python, R, SQL
- Data Engineering & Databases: DuckDB, PostgreSQL, dbt, Airflow
- Machine Learning & Modeling: Scikit-learn, Neural Networks (MLP, CNN, RNN/LSTM, Autoencoder, GAN, VAE), Logistic Regression, KNN, K-Means
- Tools: Git/GitHub, Docker, Grafana, Excel

FEATURED PROJECTS

- Youth Program Analytics & Sleep-Mood Research** • **Youth Progress** 2026
- Analyzed internal program data on youth enrollment and discharge activity, identifying key drivers of discharge outcomes; enabled program leadership to optimize resource allocation for high-risk cohorts.
 - Designed and launched a survey study on nightly sleep duration and self-reported mood; developed the initial dataset to quantify the correlation, providing actionable insights for wellness programming.
- Pretrial Detention & Conviction Outcomes in NY State** • **Capstone, Willamette University** 2026
- Joined 1.3M+ NY DCJS pretrial case records with county poverty data using DuckDB to evaluate the 2020 bail reform law's impact on conviction outcomes, employing logistic regression to control for demographic variables and prior records.
- Breast Cancer Diagnosis: Machine Learning Analysis** • **Willamette University** 2026
- Achieved 95.6% classification accuracy on the Wisconsin Breast Cancer dataset by comparing Logistic Regression and KNN, prioritizing model recall for malignant cases.
- Prague Metro Transit Data Pipeline** • **Willamette University** 2026
- Led documentation for a team data pipeline project, writing architecture decision records and validating warehouse tables; built a Grafana geomap dashboard to visualize metro ridership activity.

EDUCATION

- M.S. Data Science** • **Willamette University** *Expected August 2026*
- Graduate Coursework, Psychology & Neuroscience** • **Portland State University** *June 2025*

EXPERIENCE

- Youth Engagement Coach** • **Youth Progress** *July 2024 – Present*
- Provide BRS case management for foster youth, coaching and mentoring while maintaining 100% documentation compliance with state standards.
- Substitute Teacher** • **ESS** *October 2023 – May 2025*
- Supported individualized education plans and maintained classroom routines in special needs environments.
- Youth Advocate** • **IRCO** *January 2022 – July 2023*
- Connected immigrant and refugee youth and families to community and educational resources, facilitating successful integration.
- Case Manager, Founder of Stand With Youth** • **SACCO** *March 2021 – September 2021*
- Conducted intake assessments and developed individualized service plans for youth and families; founded a youth advocacy initiative to coordinate cross-provider referrals.